

PREVIOUSLY...

MORDELL-WEIL
(over $\mathbb{Q}$)

WEAK MORDELL-WEIL / K
DESCENT THEOREM
HEIGHTS (over $\mathbb{Q}$)

$E(\mathbb{Q}) \cong E(\mathbb{Q})_{\text{tors}} \oplus \mathbb{Z}^{R_{E/\mathbb{Q}}}$

TORSION SUBGROUP "FREE PART"

RANK OF $E(\mathbb{Q})$

TODAY: TORSION POINTS

and then canonical heights.

(VII.3.4) $\Rightarrow$ Thm. Let K be a number field, E/K an ell. curve.

w/ $y^2 + a_1xy + a_3y = \dots$

LONG WEIERS. !

such that $a_i \in \mathcal{O}_K$. Let $P \in E(K)$ be a pt of finite order $m \geq 2$.

(a) If m is not a power of a prime, then $x(P), y(P) \in \mathcal{O}_K$. (=R).

(b) If $m = p^n$, then $\forall v \in M_K^o$, let $r_v = \left\lfloor \frac{\text{ord}_v(P)}{p^n - p^{n-1}} \right\rfloor$,

$$\text{then } \text{ord}_v(x(P)) \geq -2r_v$$

$$\text{ord}_v(y(P)) \geq -3r_v$$

In particular, $x(P), y(P)$ are integral if $\text{ord}_v(P) = 0$.

Recall Over $\mathbb{Q}$: $r_q = \left\lfloor \frac{\text{ord}_q(P)}{p^n - p^{n-1}} \right\rfloor = \begin{cases} 0 & \text{if } q \neq p. \\ 0 & \text{any } n \\ & p = q > 2 \\ 1 & p = 2, n = 1 \end{cases}$

Cor. (Nagell-Lutz theorem)

$$E/\mathbb{Q} : y^2 = x^3 + Ax + B, \quad A, B \in \mathbb{Z}.$$

Suppose $P \in E(\mathbb{Q})$ is a non-zero torsion point. Then:

(a) $x(P), y(P) \in \mathbb{Z}$.

(b) Either $[2]P = \mathcal{O}$ or else $y(P)^2 \mid (4A^3 + 27B^2)$.

example $E: y^2 = x^3 - 2$ • $E[2]$ $x^3 - 2 = 0$ is irreducible over $\mathbb{Q}$
 $\Rightarrow$ no two torsion / $\mathbb{Q}$.

• $P \notin E[2], \underbrace{y(P)^2}_{\mathbb{Z}} \mid 4A^3 + 27B^2 = 27 \cdot 4$

$$\Rightarrow y \in \{\pm 1, \pm 2, \pm 3, \pm 6\}$$

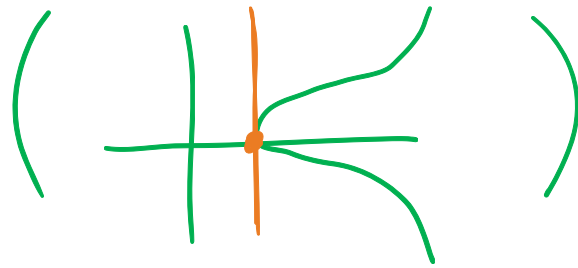
$$\Rightarrow x^3 = y^2 + 2 \in \{3, 6, 11, 38\} \cap \square = \emptyset$$

$$E(\mathbb{Q})_{\text{tors}} \cong \{\mathcal{O}\}$$

$\Rightarrow$ No Torsion over $\mathbb{Q}$!

Pf. (Nagell-Lutz) $\rightarrow$ **FOR SHORT WEIERSTRASS EQN'S !!!**

(a) If $m=2$, $y(P)=0$



$\Rightarrow x(P)$ is a root of $x^3 + Ax + B = 0$, $A, B \in \mathbb{Z} \Rightarrow x(P) \in \mathcal{O}_{\mathbb{Q}}$
 $\rightarrow$ if $x(P) \in \mathbb{Q}$, then $x(P) \in \mathbb{Z}$.

algebraic
integer

$$\hookrightarrow x(P) \in \mathcal{O}_{\mathbb{Q}} \cap \mathbb{Q} = \mathbb{Z} \quad \uparrow$$

This shows that

if $P \in E(\mathbb{Z})(\mathbb{Q}) \Rightarrow x(P), y(P) \in \mathbb{Z}$.

If $m > 2$, it follows from the Thm. $\Gamma_2 = 0$, and $x(P), y(P) \in \mathbb{Z}$.

(b) Assume $[2]P \neq \mathcal{O}$, $y(P) \neq 0$.

Then by (a), $x(P), y(P), x(2P) \in \mathbb{Z}$.

Duplication formula:
$$x(2P) = \frac{\phi(x(P))}{4\psi(x(P))}$$

where $\phi(x) = x^4 - 2Ax^2 - 8Bx + A^2$, $\psi(x) = x^3 + Ax + B$.

Lemma $\Rightarrow$
$$\underbrace{(3x^2 + 4A)}_{f(x)} \phi(x) - \underbrace{(3x^3 - 5Ax - 27B)}_{g(x)} \psi(x) = 4A^3 + 27B^2$$

• $\phi(x(P)) = x(2P) \cdot 4\psi(x(P))$, $\psi(x(P)) = y(P)^2$ ↗ plug in.

$$\Rightarrow y(P)^2 \cdot (4f(x(P)) \cdot x(2P) - g(x(P))) = 4A^3 + 27B^2$$

An eq'n of integers! $\Rightarrow y(P)^2 \mid 4A^3 + 27B^2$

Thm (Mazur, 1977 - Conjectured: Levi, Ogg)

$$E(\mathbb{Q})_{\text{tors}} \cong \begin{cases} \mathbb{Z}/N\mathbb{Z} & , 1 \leq N \leq 10 \text{ or } 12, \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^M\mathbb{Z} & , 1 \leq M \leq 4. \end{cases}$$

Recall: Weil pairing $\rightarrow$ if $E[m] \subseteq E(K)$, then $\mu_m \subseteq K$.

Fact: Every subgp in Mazur's list occurs for ∞ 'ly many j -invariants over $\mathbb{Q}$.

example Ell. curves / $\mathbb{Q}$ w/ $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/7\mathbb{Z}$ belong to a 1-parameter family:

$$y^2 + (1+d-d^2)xy + (d^2-d^3)y = x^3 + (d^2-d^3)x^2$$

with $P = (0,0) \in E[7]$, $d \in \mathbb{Q}$, $(\Delta \neq 0)$.

Reference: Weston's "The modular curves $X_0(11)$ and $X_1(11)$ " $X_1(7)$

CANONICAL HEIGHTS

$$H: \mathbb{Q} \rightarrow \mathbb{R}, \quad H(P/Q) = \max \{ |P|, |Q| \}$$

$$h_X: E(\mathbb{Q}) \rightarrow \mathbb{R}, \quad h_X(P) = \begin{cases} \log H(X(P)) & \text{if } P \neq \mathcal{O}, \\ 0 & \text{if } P = \mathcal{O}. \end{cases}$$

In general, if f is an even function on $E(\mathbb{Q})$,

$$\text{define } h_f(P) = \begin{cases} \log H(f(P)) & \text{if } P \neq \mathcal{O}, \\ 0 & \text{if } P = \mathcal{O}. \end{cases} \quad \left. \vphantom{\begin{matrix} \log H(f(P)) \\ 0 \end{matrix}} \right\} \text{ also a height!}$$

Thm $E/\mathbb{Q}$, $f \in \mathbb{Q}(E)$ is even ($f(P) = f(-P)$). Then

$$\bullet \forall P, Q \in E(\mathbb{Q}) : h_f(P+Q) + h_f(P-Q) = 2h_f(P) + 2h_f(Q) + O(1)$$

(i.e., h_f is "almost" a quad. form)

$$\bullet f, g \in \mathbb{Q}(E) \text{ are even, then } (\deg g)h_f = (\deg f)h_g + O(1)$$

constant depends
only on
 E and f .

Prop $E/\mathbb{Q}$, $f \in \mathbb{Q}(E)$ even, $P \in E(\mathbb{Q})$. Then the limit:

$$\frac{1}{\deg f} \lim_{N \rightarrow \infty} \frac{h_f([2^N]P)}{4^N} \text{ exists and it's independent of } f.$$

Pf. • Show that seq. is Cauchy.

(b) heights ($h_f(2P) \geq 4h_f(P) - C_2$) $\Rightarrow \exists C \cdot \forall Q \in E(\mathbb{Q}) \quad |h_f([2]Q) - 4h_f(Q)| \leq C$

Let $N \geq M \geq 0$ be integers,

$$\begin{aligned} \left| \frac{h_f([2^N]P)}{4^N} - \frac{h_f([2^M]P)}{4^M} \right| &= \left| \sum_{n=M}^{N-1} \frac{h_f([2^{n+1}]P)}{4^{n+1}} - \frac{h_f([2^n]P)}{4^n} \right| \\ &\leq \sum_{n=M}^{N-1} \frac{|h_f([2^{n+1}]P) - 4h_f([2^n]P)|}{4^{n+1}} \stackrel{Q=[2^n]P}{\leq} \sum_{n=M}^{N-1} \frac{C}{4^{n+1}} \leq \frac{C}{3 \cdot 4^M} \rightarrow 0 \text{ as } N \geq M \geq 0, M \rightarrow \infty \\ &\Rightarrow \text{Cauchy!} \end{aligned}$$

Ind.) If g is also even, $(\deg g) h_f = (\deg f) h_g + O(1)$

$$\frac{1}{\deg g \deg f} \left((\deg g) \frac{h_g([2^N]P)}{4^N} - (\deg f) \frac{h_f([2^N]P)}{4^N} \right) = \frac{1}{\deg g \deg f} O(1) \xrightarrow{\text{as } N \rightarrow \infty} 0$$

$$\lim_{N \rightarrow \infty} \frac{1}{\deg f} \frac{h_f}{4^N} - \lim_{N \rightarrow \infty} \frac{1}{\deg g} \frac{h_g}{4^N} = 0$$

so limits coincide regardless of even fn. ▢

Def. The canonical Néron-Tate height on $E(\mathbb{Q})$ is

$$\hat{h} : E(\mathbb{Q}) \rightarrow \mathbb{R}, \quad \hat{h}(P) = \frac{1}{\deg f} \lim_{N \rightarrow \infty} \frac{h_f([2^N]P)}{4^N}$$

where f is an arbitrary even fn.

(NOTE: $\hat{h}(O) = 0$.)

Thm. $E/\mathbb{Q}$, with can. height $\hat{h}$.

(e) $f \in \mathbb{Q}(E)$ even, then $(\deg f) \cdot \hat{h} = h_f + O(1)$ constant depends only on E, f .

(a) $\forall P, Q \in E(\mathbb{Q})$

$$\hat{h}(P+Q) + \hat{h}(P-Q) = 2\hat{h}(P) + 2\hat{h}(Q) \quad (\text{"parallelogram law"})$$

(b) $\forall P \in E(\mathbb{Q}), m \in \mathbb{Z}, \hat{h}([m]P) = m^2 \hat{h}(P)$.

(c) $\hat{h}$ is a quad. form on E , i.e., $\hat{h}$ is even and

$$\langle, \rangle : E(\mathbb{Q}) \times E(\mathbb{Q}) \longrightarrow \mathbb{R}$$

$(P, Q) \longmapsto \langle P, Q \rangle = \hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q)$ is bilinear.

(d) $P \in E(\mathbb{Q}), \hat{h}(P) \geq 0$ and $\hat{h}(P) = 0 \iff P$ is torsion. (!! useful !!)

If $\hat{h}'$ satisfies (e) and (b) for any $m \geq 2$, then $\hat{h}' = \hat{h}$.

